ECHO Magnet

Extremely Curved Hadrontherapy Optics Magnet

Proposal for young researchers grant – CSN5

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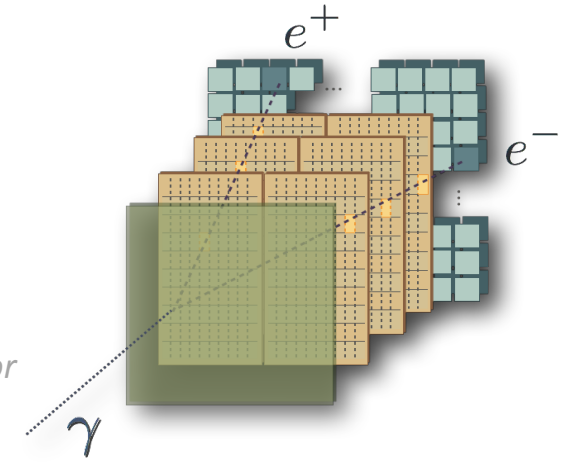
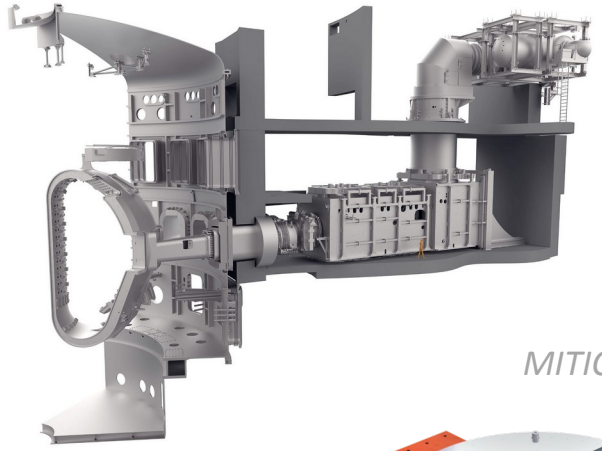
About me:

- 2020 – Bachelor degree in Physics
  - University of Milan, Medical Physics
    ”Feasibility study of an innovative range monitoring technique using prompt gamma in hadrontherapy”
- 2022 – Master degree in Fusion Plasma Physics
  - University of Milan, CNR Padova – Consorzio RFX
    ”Set-up of the spectroscopic diagnostics for MITICA”
- 2022 – Present PhD in Applied Physics (Final defense expected early jan 2026)
  - University of Milan, LASA laboratory
    ”Design and study of the IRIS 10 T Energy Saving dipole Magnet for Accelerators ESMA”

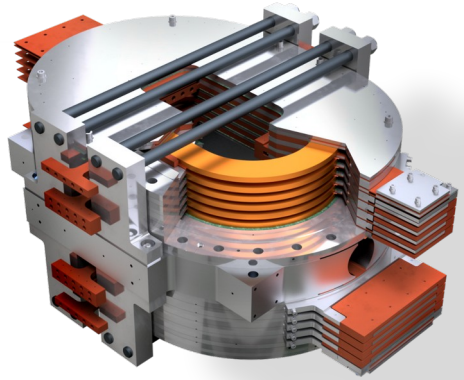
→ **CSC Graduate Study Fellowship Award by IEEE, Council on SuperConductivity (CSC)**



PAPRICA detector



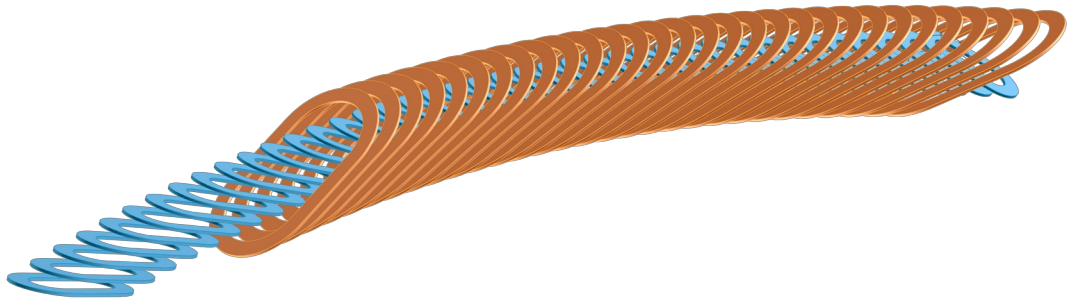
MITICA Neutral Beam Injector



ESMA dipole magnet

Outline

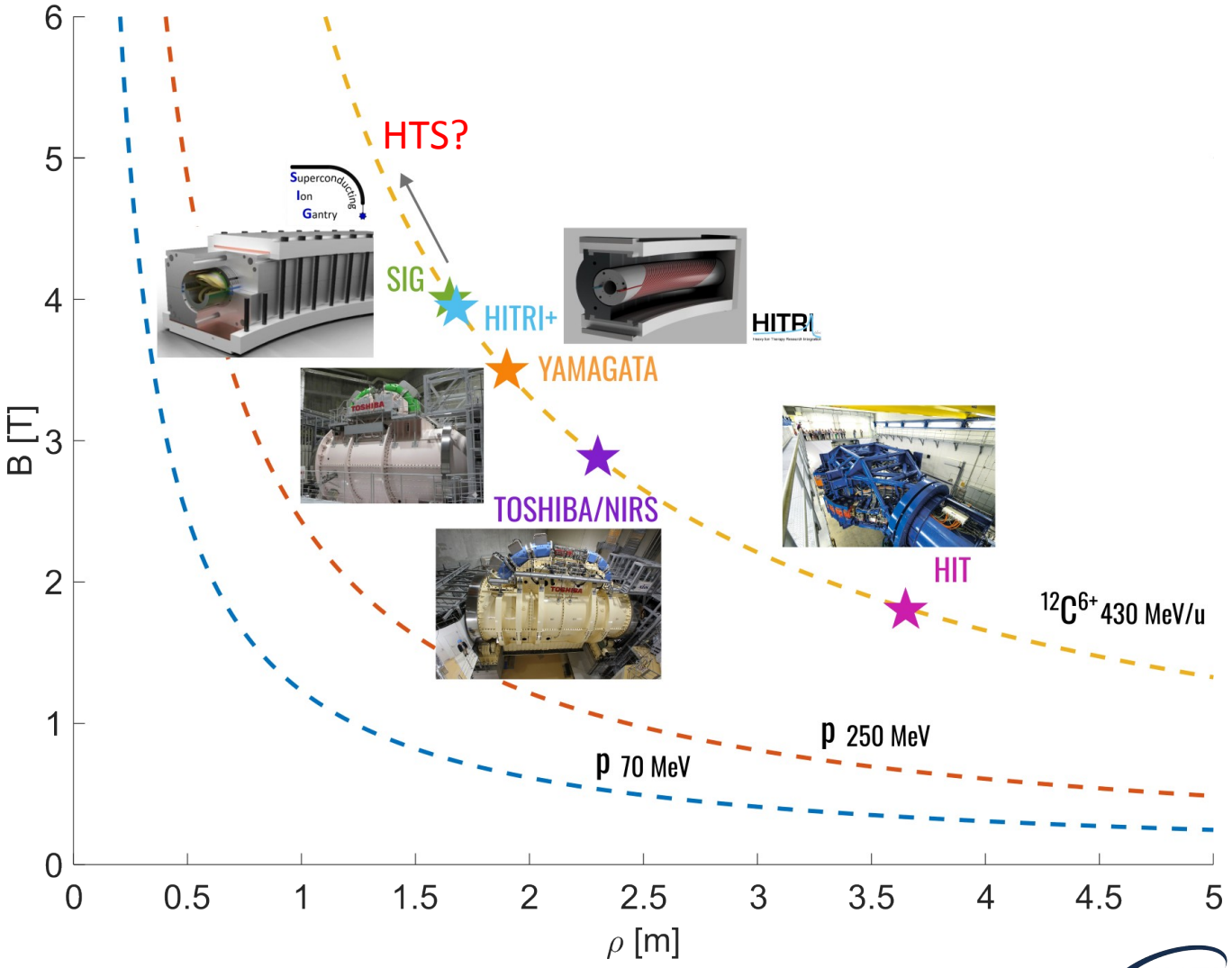
- Status of the art,
  Why ECHO Magnet?
- Realization plan of ECHO
  1. Design
  2. Manufacturing
- Timeline, risks and costs analysis
- Conclusions



State-of-the-art & actual R&D

- Resistive magnets for ion gantry:
  - HIT: B = 1.8 T, ρ = 3.65 m
  - 650 tons
- Low Temperature Superconductors (LTS)
  - Toshiba/NIRST: B = 2.88 T, ρ = 2.3 m,
    - 300 tons
  - Yamagata: B = 3.5 T, ρ = 1.9 m
    - 200 tons
  - SIG: B = 4 T, ρ = 1.65 m
  - HITRI+: B = 4 T, ρ = 1.65 m

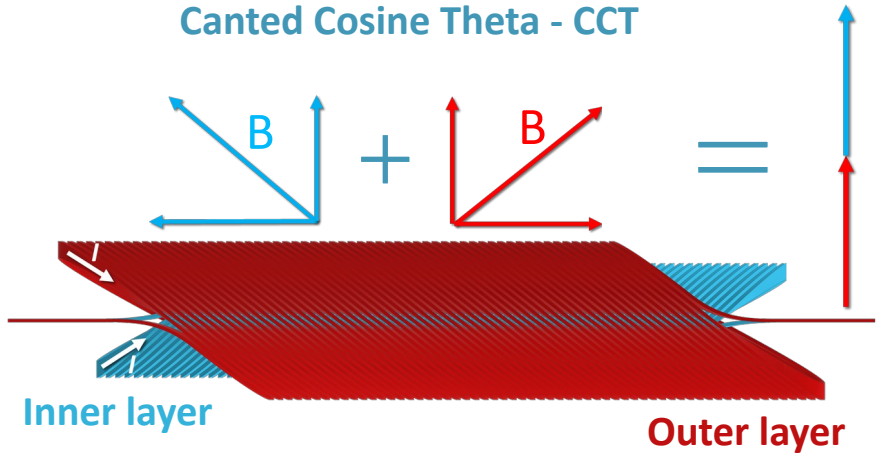
  } m ~ 100 tons
- High Temperature Superconducting (HTS) tapes are promising candidates

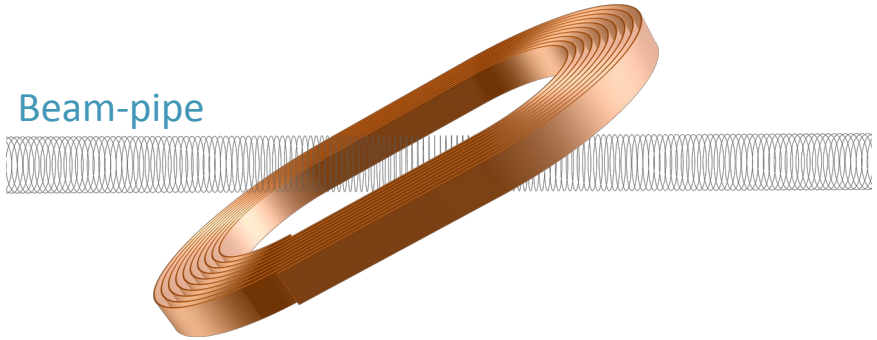


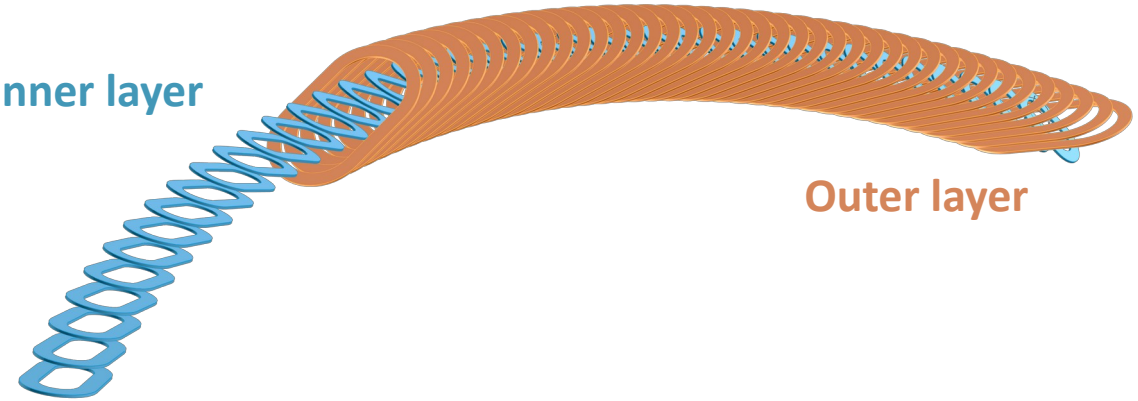
ECHO Magnet: Targets

Extremly Curved Hadrontherapy Optics (ECHO) Magnet:

- Gantries and particle accelerators for hadrontherapy.
- Demonstrating the **T**echnological **R**eadiness **L**evel (TRL) of **H**igh **T**emperature Superconductors (HTS)
- Energy saving with respect to classical **L**ow **T**emperature **S**uperconductors (LTS)
- Modular design assembled in a curved geometry







Target central field B_0	T	**5**
Curvature radius	m	**1.3**
Free aperture ∅	mm	50
Operating temperature	K	20
Critical current margin	N/A	>20%

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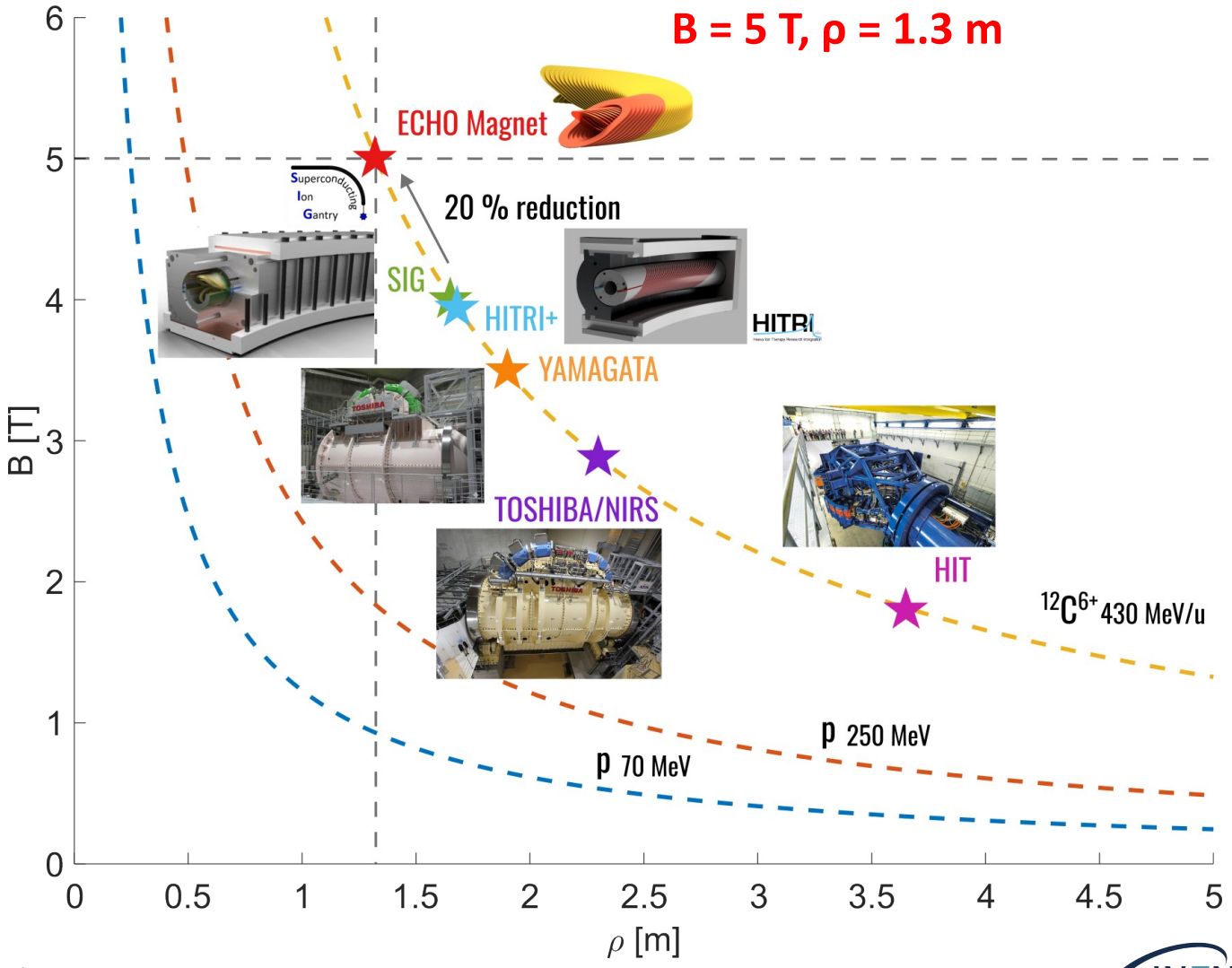
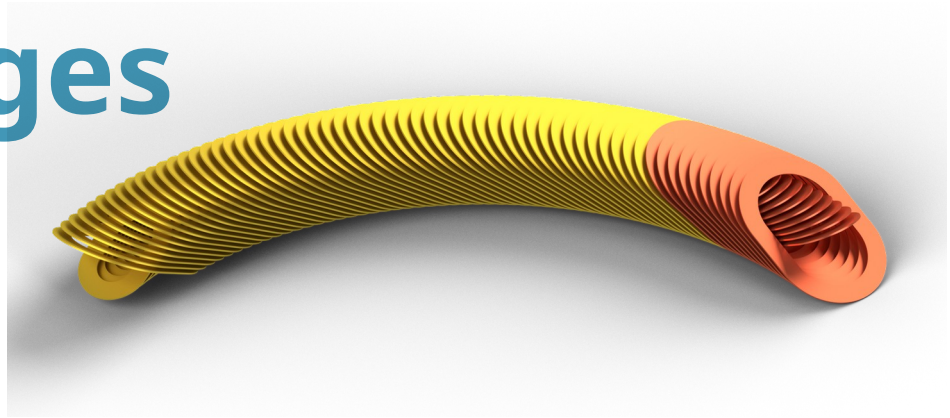
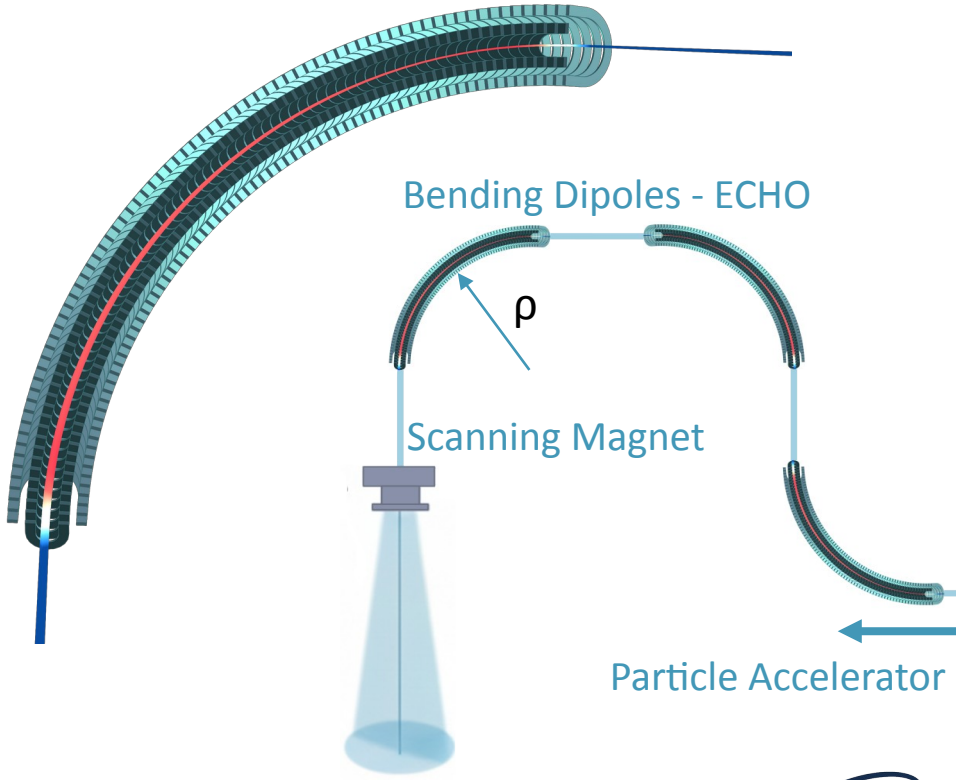


ECHO Magnet: Advantages

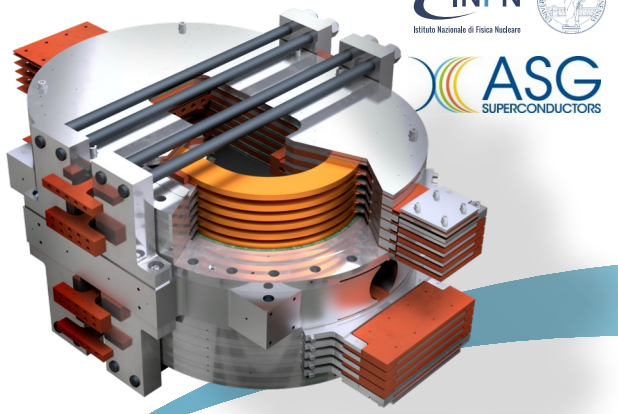


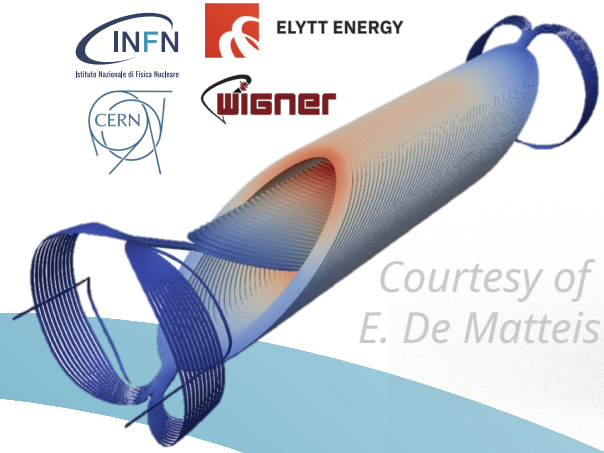
Given the state of the art on SC dipole magnets for hadrontherapy ECHO magnet introduces:

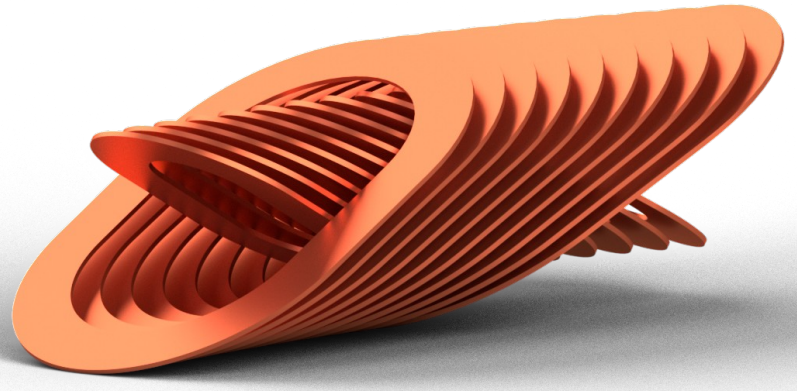
- Operation at 20 K liquid He free
  - x5 energy saving factor with respect to 4.2 K (just for COP)
- Modular design: industrialization and scalability
- Target central field of 5 T
  - 20 % reduction of the curvature radius **ρ**
- QA procedure at 77 K granted by HTS



Dipolar HTS magnet towards ECHO







- IRIS dipole - ESMA

Pros
- Flat coils → windability
- High field (10 T)
- Solid design

Cons
- Straight geometry
- ± 1.5 % field quality

- I.FAST CCT dipole

Pros
- Field quality
- No inner junctions

Cons
- Straight geometry
- Complex helix-winding
- Critical external junctions

- **ECHO Magnet**

Pros
- Flat coils → windability
- Curved geometry
- Field quality
- Scalability

Cons
- Modules assembly on curved support
- Resistive joints

Roadmap of activities for ECHO

Phase 1	Characterization of HTS tape	• Electrical • Thermal and Mechanical
	Single module design and challenges	
Phase 2	Modules manufacturing	
	Mechanical Assembly	
	Final Testing	

Roadmap of activities for ECHO

Phase 1

- Characterization of HTS tape
  - Electrical
  - Thermal and Mechanical
- Single module design and challenges

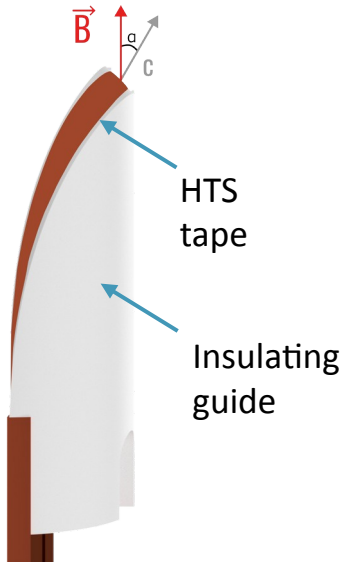
Phase 2

- Modules manufacturing
- Mechanical Assembly
- Final Testing

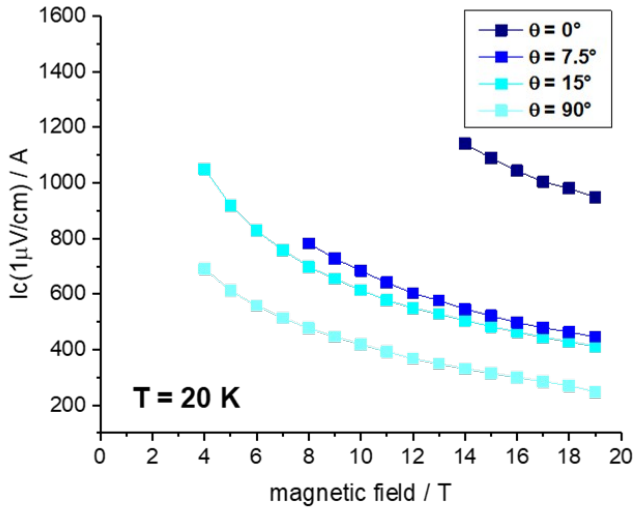
Phase 1: Characterization of HTS tape
Electrical

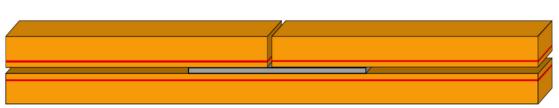
Electrical properties are fundamental to drive the electromagnetic design of each module:

- HTS tape critical current dependencies:
  - Temperature
  - Magnetic field
  - Angular
- Optimization of resistive splices
  1. Within each module
  2. Inter-module

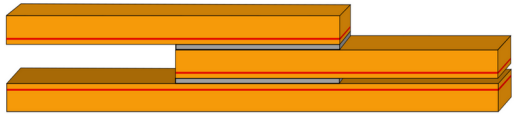


Courtesy of C. Senatore, University of Geneva

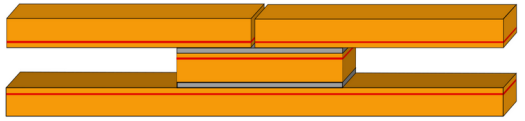




Simple Flat



Lap



Bridge

Joint Failure

Roadmap of activities for ECHO

Phase 1

- Characterization of HTS tape
  - Electrical
  - Thermal and Mechanical
- Single module design and challenges

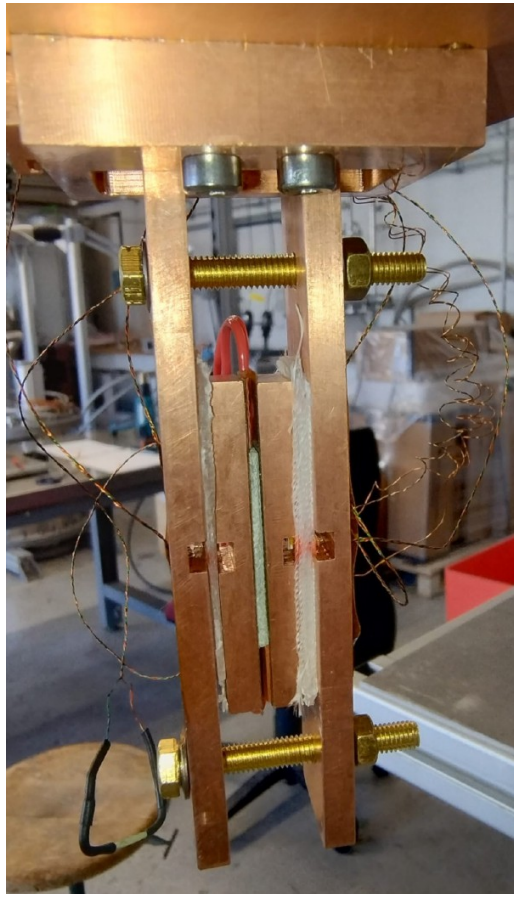
Phase 2

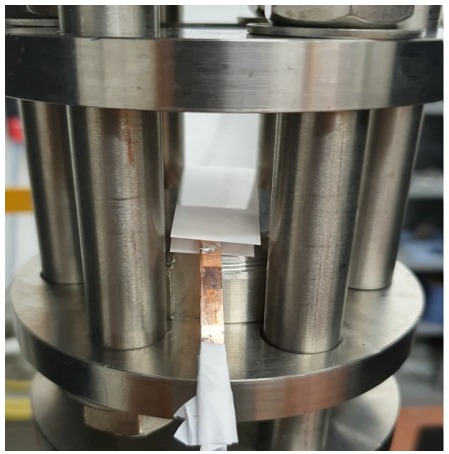
- Modules manufacturing
- Mechanical Assembly
- Final Testing

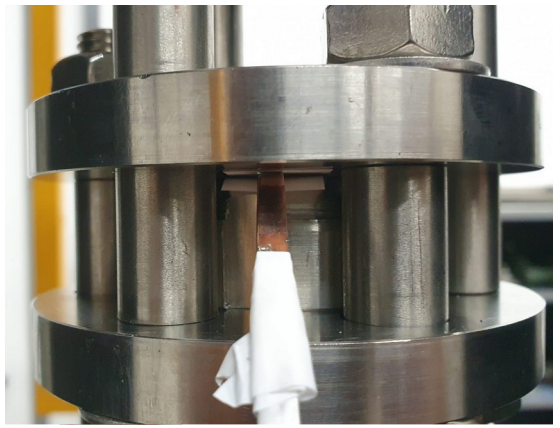
Phase 1: Characterization of HTS tape
Thermal and Mechanical

Thermal and mechanical properties will be studied for different cables configurations:

- Thermal conductivity
  - Dry-wound
  - Impregnated
- Lower than expected Young's modulus
  - Dry-stack
  - Impregnated stack







Roadmap of activities for ECHO

Phase 1

Characterization of HTS tape
- Electrical
- Thermal and Mechanical

Single module design and challenges

Phase 2

Modules manufacturing

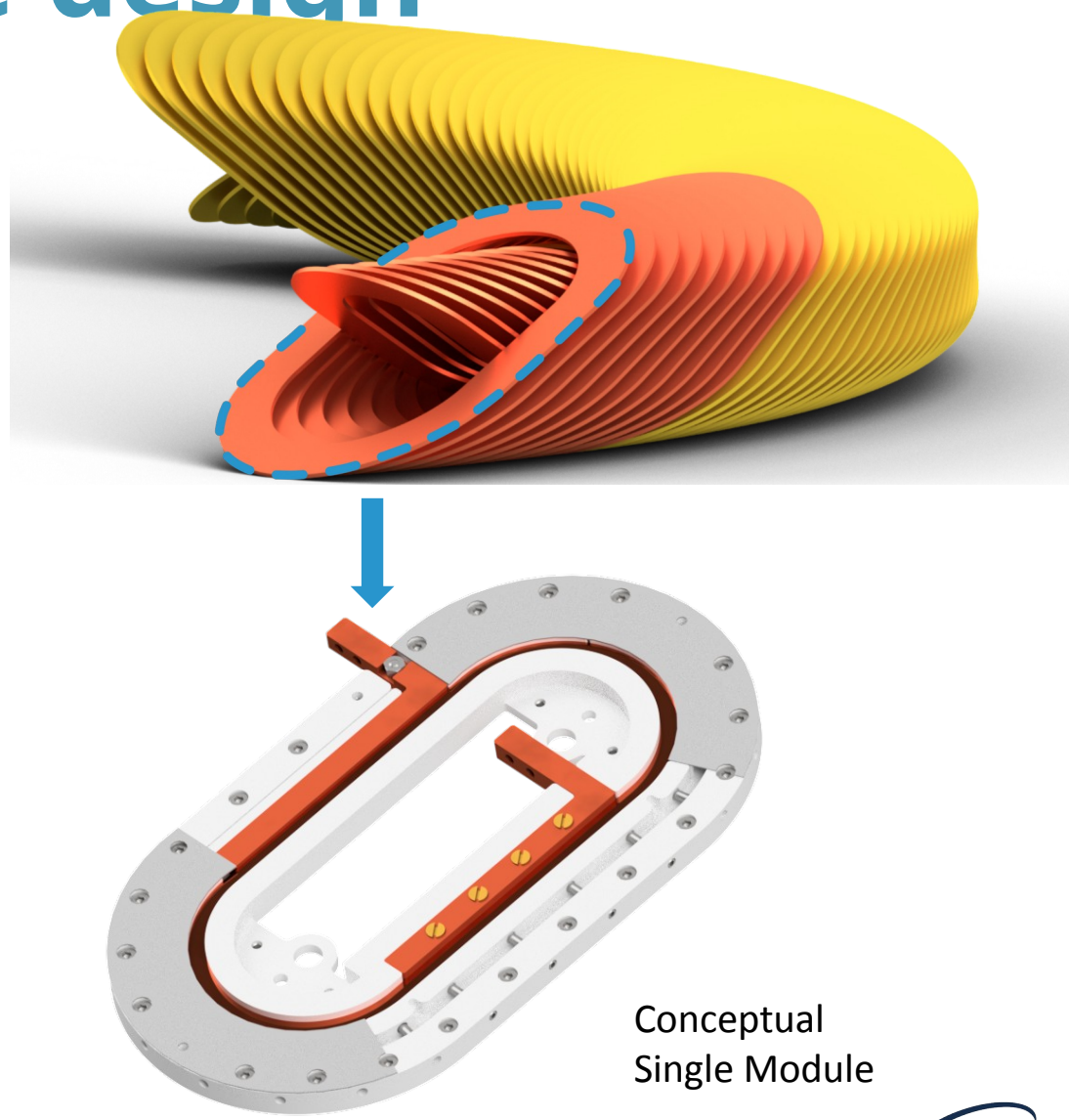
Mechanical Assembly

Final Testing

Phase 1: Single module design

A small HTS racetrack coil design aiming at one ECHO module will be performed.
The design must assess the following question:

- Realize a solid design with operational margins
- Achieve a repeatible field quality
- Optimize resistive joints
- Handle the assembly on a curved support
- Set up a QA procedure in LN_2
  - ➢ Modularity and repeatibility of the process



Conceptual
Single Module

Roadmap of activities for ECHO

Phase 1

- Characterization of HTS tape
  - Electrical
  - Thermal and Mechanical
- Single module design and challenges

Phase 2

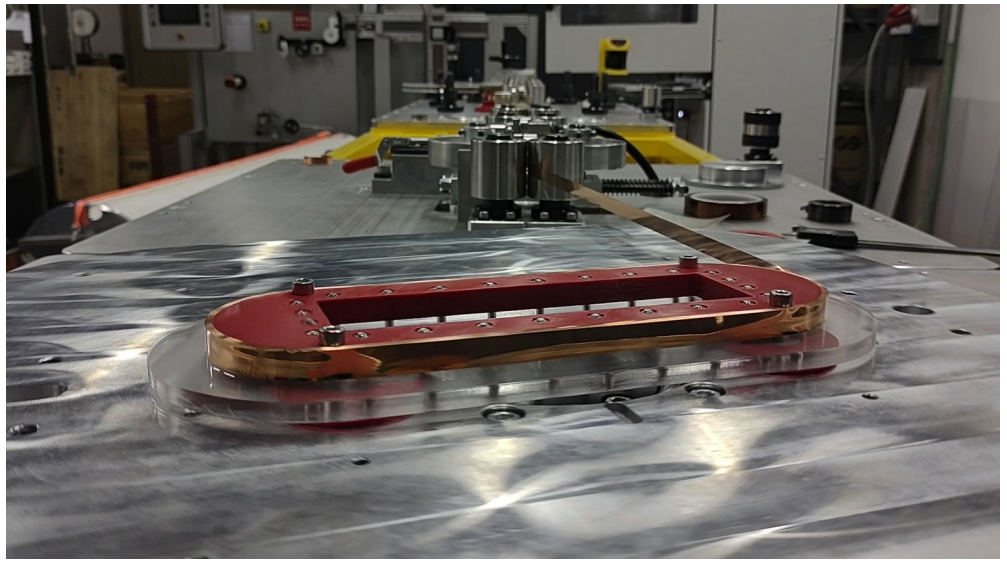
- Modules manufacturing
- Mechanical Assembly
- Final Testing

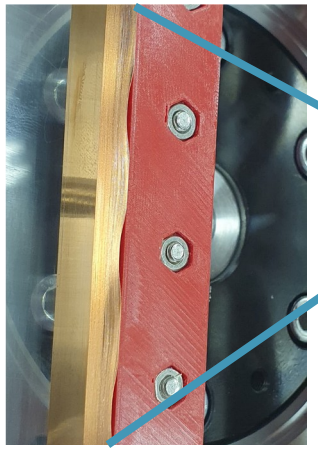
Phase 2: Module Manufacturing

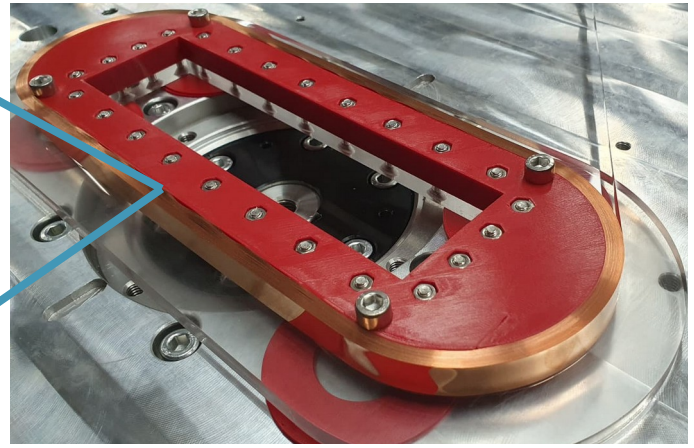
The winding of each ECHO module will be carried out considering:

- Different winding technologies using a multi-tape winding machine:
  - Up to 6 tapes in parallel.
  - Variable and controlled winding tension.
- A classical racetrack geometry suffers from buckling:
  - Alternatives/mitigations will be studied.

Winding test: single tape on plastic mandrel







Roadmap of activities for ECHO

Phase 1

Characterization of HTS tape
- Electrical
- Thermal and Mechanical

Single module design and challenges

Phase 2

Modules manufacturing

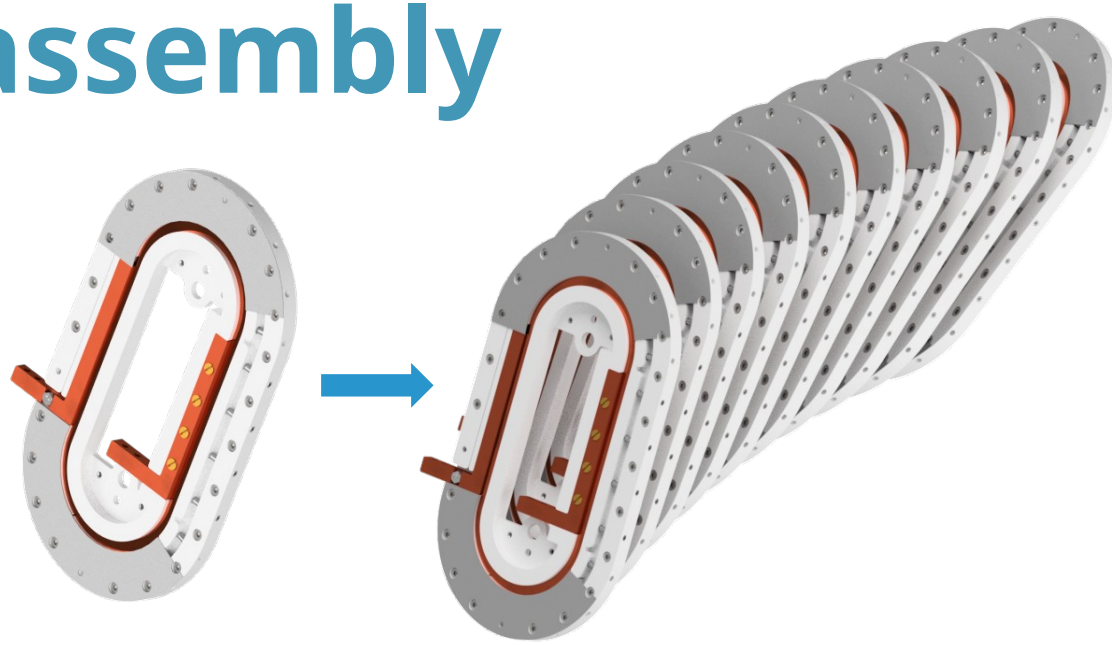
Mechanical Assembly

Final Testing

Phase 2: Mechanical assembly

The manufacturing of the ECHO module will be carried out considering:

- Integrating the mechanic of each module with the curved support
- Optimization of resistive junctions between modules
- Possibility to exploit metallic additive manufacturing:
  - Copper
  - Stainless steel



Roadmap of activities for ECHO

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- Characterization of HTS tape
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Phase 2

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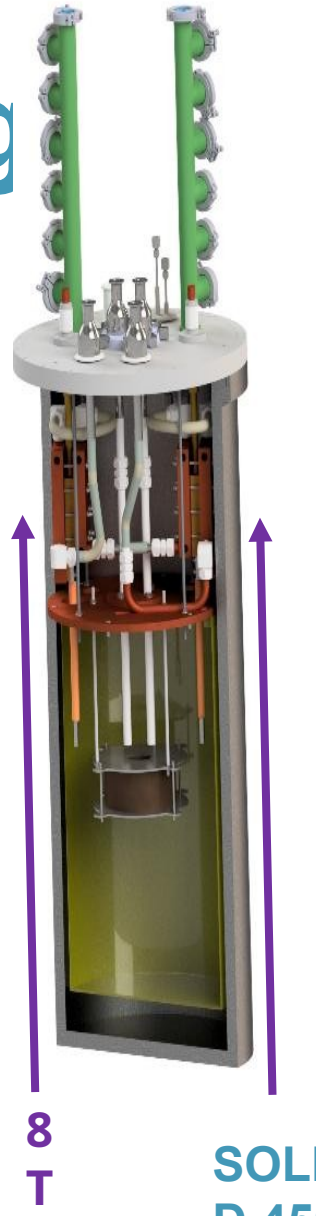
- Mechanical Assembly
- Final Testing

Phase 2: Final testing

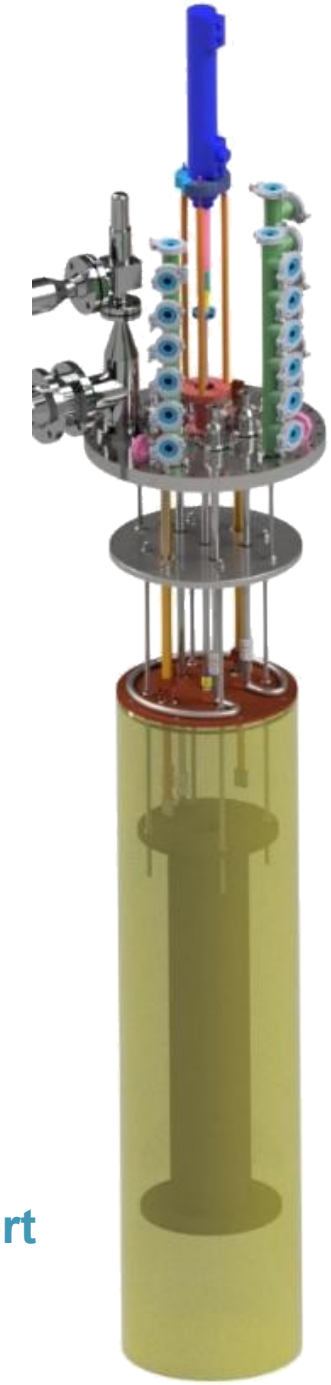
Tests at 20 K will be carried out inside the IRIS variable temperature test stand for HTS magnets:

Closed loop cooling 20 K – 50 K

- Based on cryocoolers and closed loop G-Helium :
  - 20 W @ 20 K
    (Adjustable T from 20 K to 50 K)
  - 300 W @ 55 K



SOLEMI insert
D 450 mm
h 720 mm

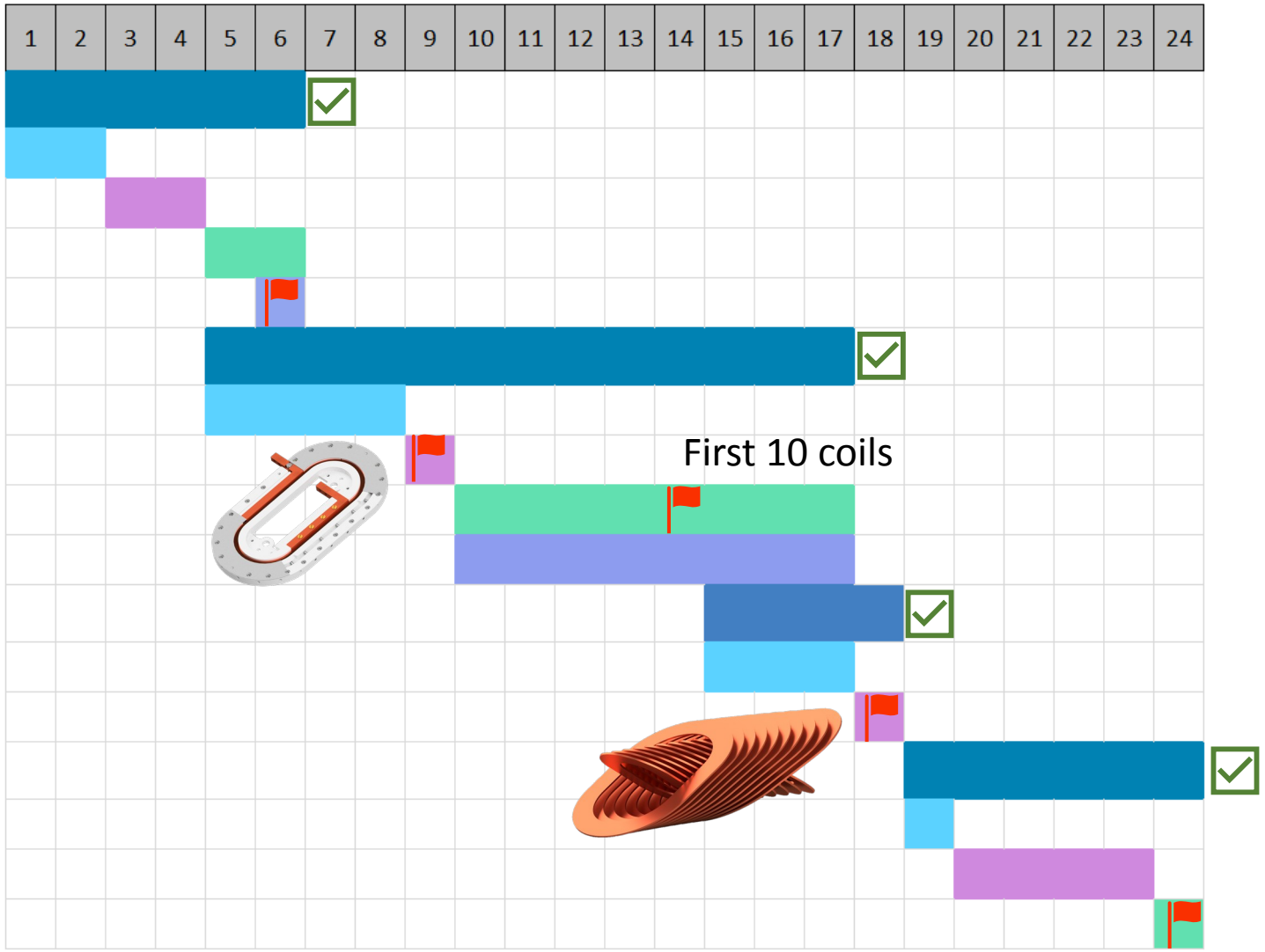


HiLumi
D 450 mm
h 1910 mm

Project timeline

Milestones:
Deliverables:

Task description	Starting	Expected duration
Magnet Design	1	6
Electromagnetic design	1	2
Mechanical design	3	2
Thermal Design	5	2
Design assesment	6	1
Coil Fabrication and testing	5	13
First prototype realization	5	4
First prototype test	9	1
Coil manufacturing	10	8
Coil testing	10	8
Assembly and Integration	15	4
Support manufacturing	15	3
Assembly	18	1
Cryogenic test	19	6
Assesment of the assembly	19	1
Testing preparation	20	4
Final test	24	1



Risks and costs summary

About the risk analysis:

- Risks are mainly calculated from the experience accumulated in the design and manufacturing of ESMA
- Windability may be affected by the chosen geometry and superficial homogeneity of the tape
- Protection and thermal budget must be carefully addressed, as well as field quality

Risk Matrix		Impact				
		Insignificant	Minor	Moderate	Major	Severe
Likelihood	Almost Certain	Medium	High	Very High	Very High	**Field Quality** Very High
	Likely	Medium	**HTS defects** High	High	**Protection** Very High	Very High
	Possible	Low	Medium	High	High	Very High
	Unlikelyh	Low	Low	**Test issues** Medium	Medium	High
	Rare	Low	Low	Low	**Windability** Low	Medium

Risks and costs summary

Main mitigations:

- **Field quality**: in the design phase consider a narrow tape (4 mm, or less).
- **Protection**: pass from insulated to Controlled Resistance (CI) mode.
- **Defects**: cut the weak portion of the cable and optimize joint procedure.
- **Test issues**: Increase the cooling power with a suitable additional variable temperature heat exchanger.
- **Windability**: Ask for more demanding thickness specification to the manufacturerand act on the design of the flat coil geometry.

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Likelihood	Almost Certain	Medium	High	Very High	Very High	Very High
	Likely	HTS defects Medium	High	Field Quality High	Very High	Very High
	Possible	Low	Medium	Protection High	High	Very High
	Unlikelyh	Low	Test issues Low	Medium	Medium	High
	Rare	Low	Low	Windability Low	Low	Medium

Risks and costs summary

Costs analysis considered:

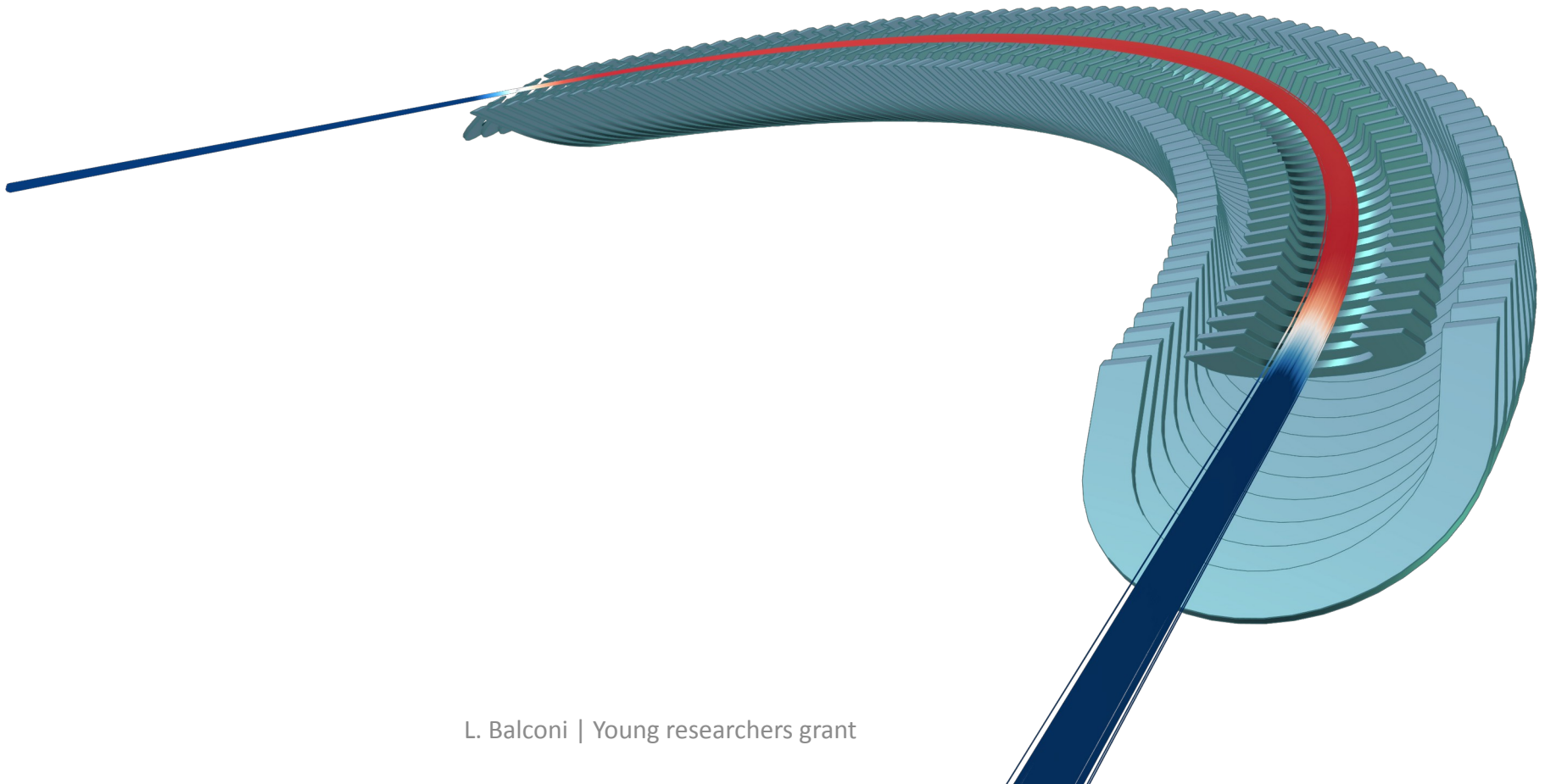
- Tape quotation was obtained addressing medium performance, commerically available 4 mm HTS tape
- Mechanical components for each coil were based on a preliminary design for small HTS racetrack coil
- Curved mechanical supports and cryogenic tests component are based on SIG mechanics and from the preparation of the test of a single ESMA coil

Object	Year	Cost [€]	Cost + IVA [€]
3 km of HTS tape	1+2	55 500	67 710
Coils mechanical components	1	22 800	27 816
Curved mechanical supports	2	12 250	14 945
Cryogenic test components	2	25 300	30 866
Presentation at conferences	1+2	3 600	3 600
Total	**1+2**	**119 450**	**144 937**

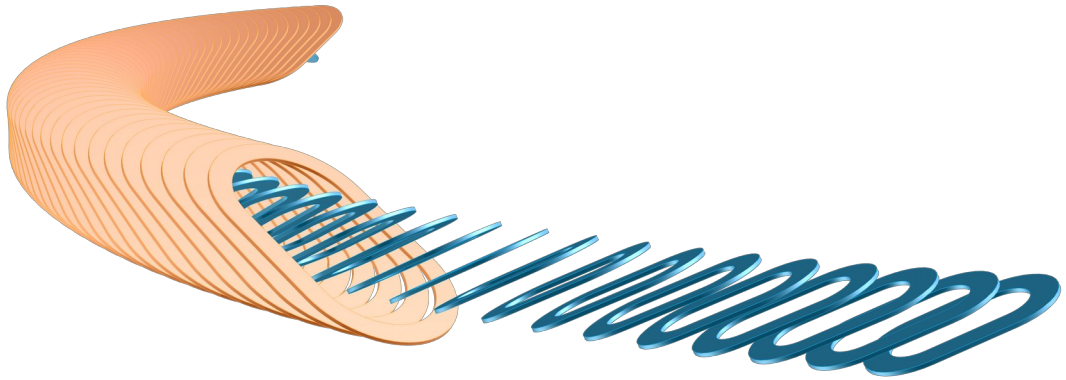
Final recap and conclusions

- Given the expertise gained from **INFN-led** project as **SIG** e **HITRI+** on SC gantry, a **classical CCT** design was adapted to be realized in a **modular structure**:
  - The goal is to realize a **small-scale HTS** dipole magnet demonstrator consisting of **10 modules** for each layer.
- The project sets the **first milestone** towards a **next generation** of ion gantries introducing:
  - Energy saving.
  - Industrial scalability.
  - Compact gantries.

The end



Backup Slides

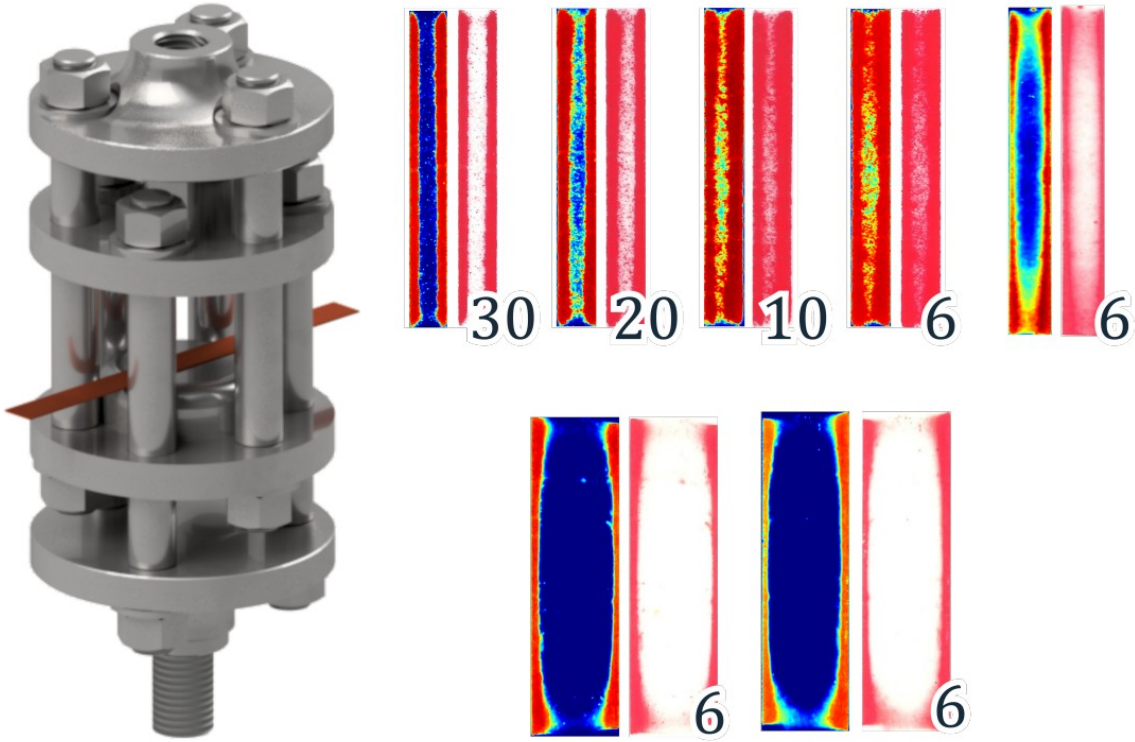


Thickness Inhomogeneity

Tapes may show thickness inhomogeneity commonly referred to as “dog-bone”, occurring during the electroplating of the final copper layer.

This may lead to:

- Localized stresses
- Winding issues



Supplier Spec 67 µm ± 10% (from 59 µm to 74 µm

Inclusion Characterisation

64 Spools fully de-spooled and inspected for defects

Minimum Inclusion thickness tracked: 30 µm

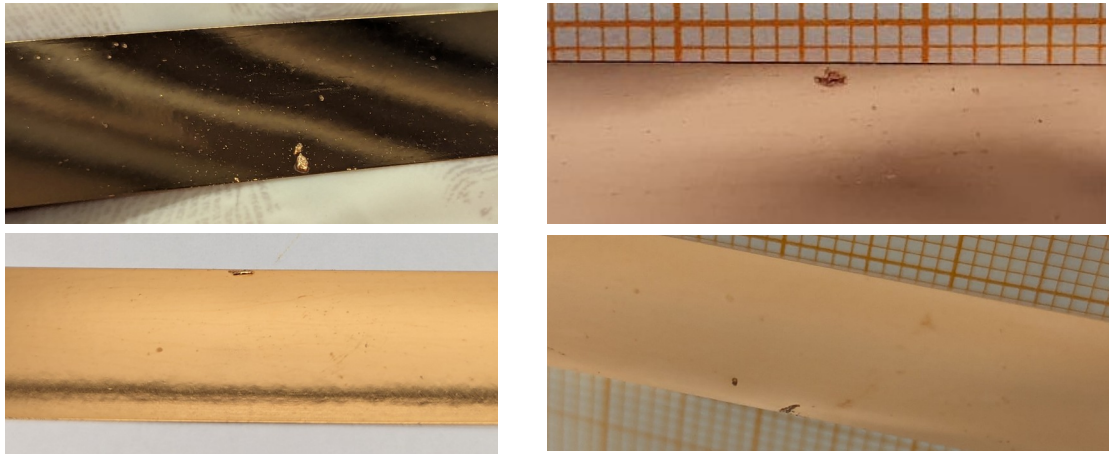
Inclusions: > 400

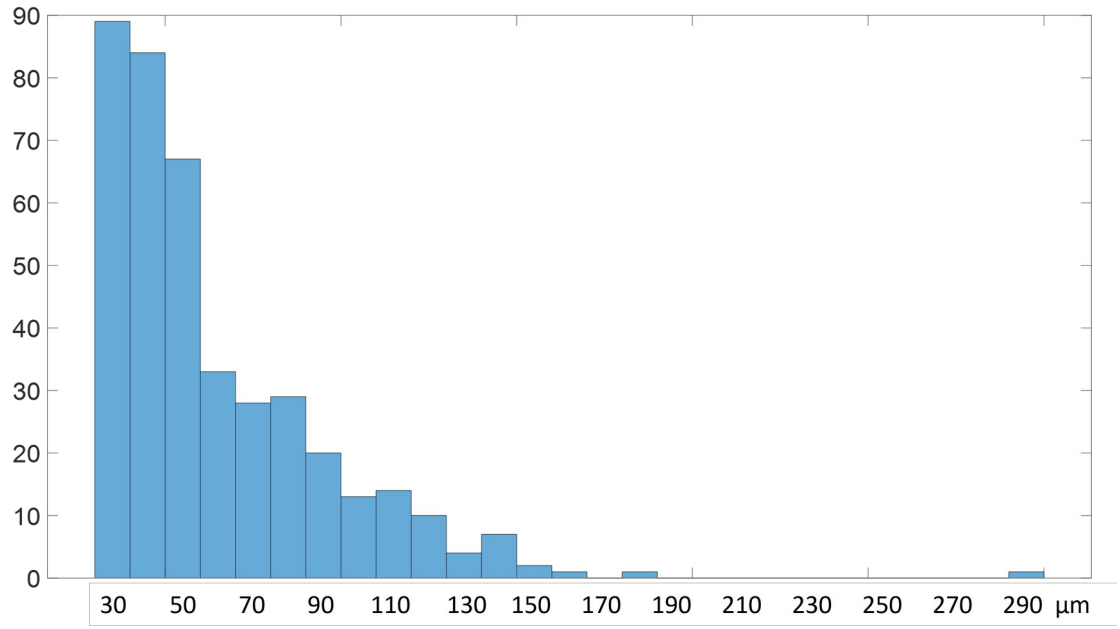
95% Inclusion on the good side (YBCO side)

80% Inclusion on the edge areas

Inclusion Genesis – Silver Sputtering

1. small silver particles can stick to the rollers in the tape transport system
2. they stick to the tape surface
3. they get coated with copper during electroplating

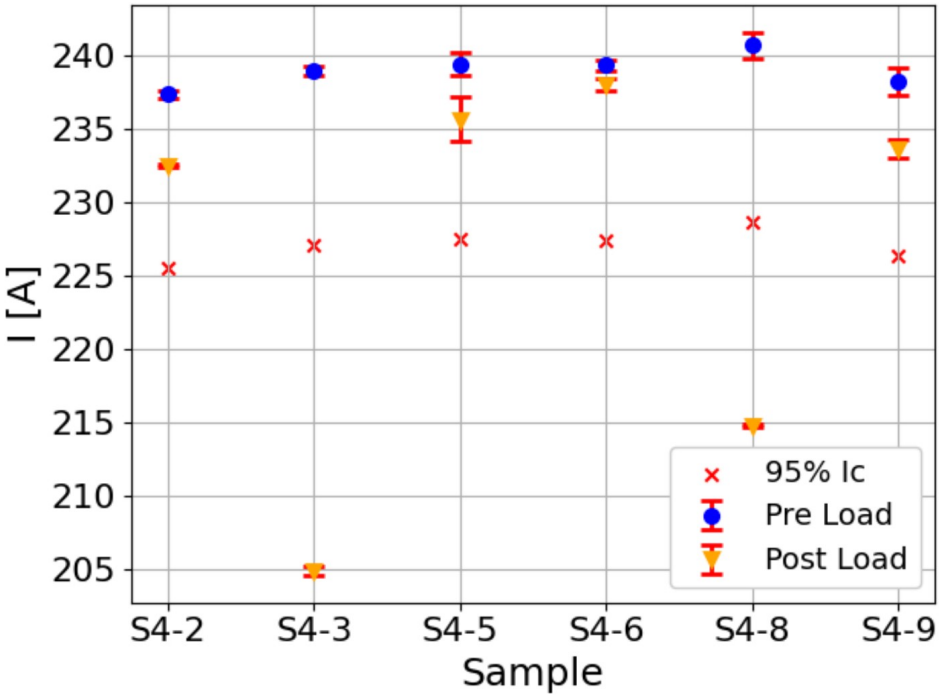




Characterization of HTS stacks

Mechanical measurement will be followed by electrical measurements to assess a possible tape degradation:

- Variable uniaxial compressive load
- Different stack conditions



Resources availability

Endorsement of the magnet group leader at LASA for the use of the following resources:

- Group technicians for coil assembly and manufacturing up to 2 FTE